Speeding up spicy-snow S1 downloads

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Methods tested

- Three pipelines for downloading Sentinel-1 radiometrically corrected datasets were tested.
- **Hyp3** – the current pipeline. Uses gamma processors running on AWS nodes to do real-time coregistration and terrain corrections.
- **Opera RTC** – near-global near-real time JPL project to generate coregistered Sentinel-1 RTC products.
- **Microsoft Stac** – Stac catalog hosted by Microsoft of Sentinel-1 RTCs

Hyp3

- Pros
  - Lots of control over resolution, dem used, processing choices
  - Currently implemented
- Cons
  - Limited by Hyp3 quota per month
  - Very slow to run the cloud pipeline and download full SLCs

Opera

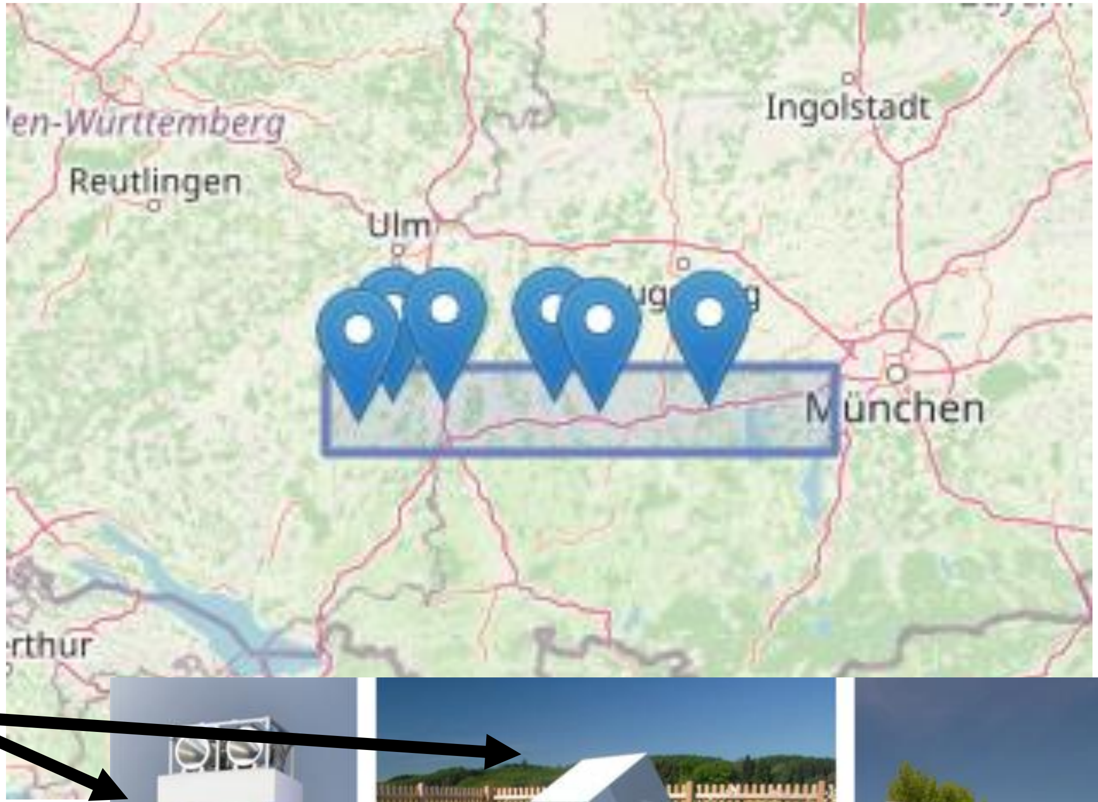
- Pros
  - Already processed so no cloud processing time
  - No cost to download
  - JPL is trusted source
- Cons
  - Smaller tiles means lots of small files to download
  - Not a stac catalog

Microsoft

- Pros
  - Super fast
  - Can clip directly from AOI into xarray dataset
- Cons
  - Uncertain of Microsoft's SAR capabilities in terms of radiometric corrections or co-registration

Test AOI and date selection

- Running over the CR reflector sites at DLR
- Blue dots represent corner reflectors (either mechanic left image or transponders center and left images).
- Ran over last 12 months = ['2024-11-11', '2025-11-11’]



Transponders are targets 3- 6

Movable passive targets are 1-3

Processing time selection (getting all dates)

Testing whether all overpass dates are present

- Hyp3 – 45 dates found. First 2024-11-22, last 2025-11-05

- Opera – 43 dates found. First 2024-11-22, last 2025-11-05. Missing 2025/05/03 and 2025/05/15 relative to hyp3

- Microsoft Stac – 47 dates found, first 2024-11-22, last 2025-11-11.

Got the overpass directly on day of process (2025-11-11). Has one extra day on 2025-03-28 that is entirely -32768 values…

Processing time tests

These represent time from feeding dates and AOI to getting an xarray dataset of vv and vh

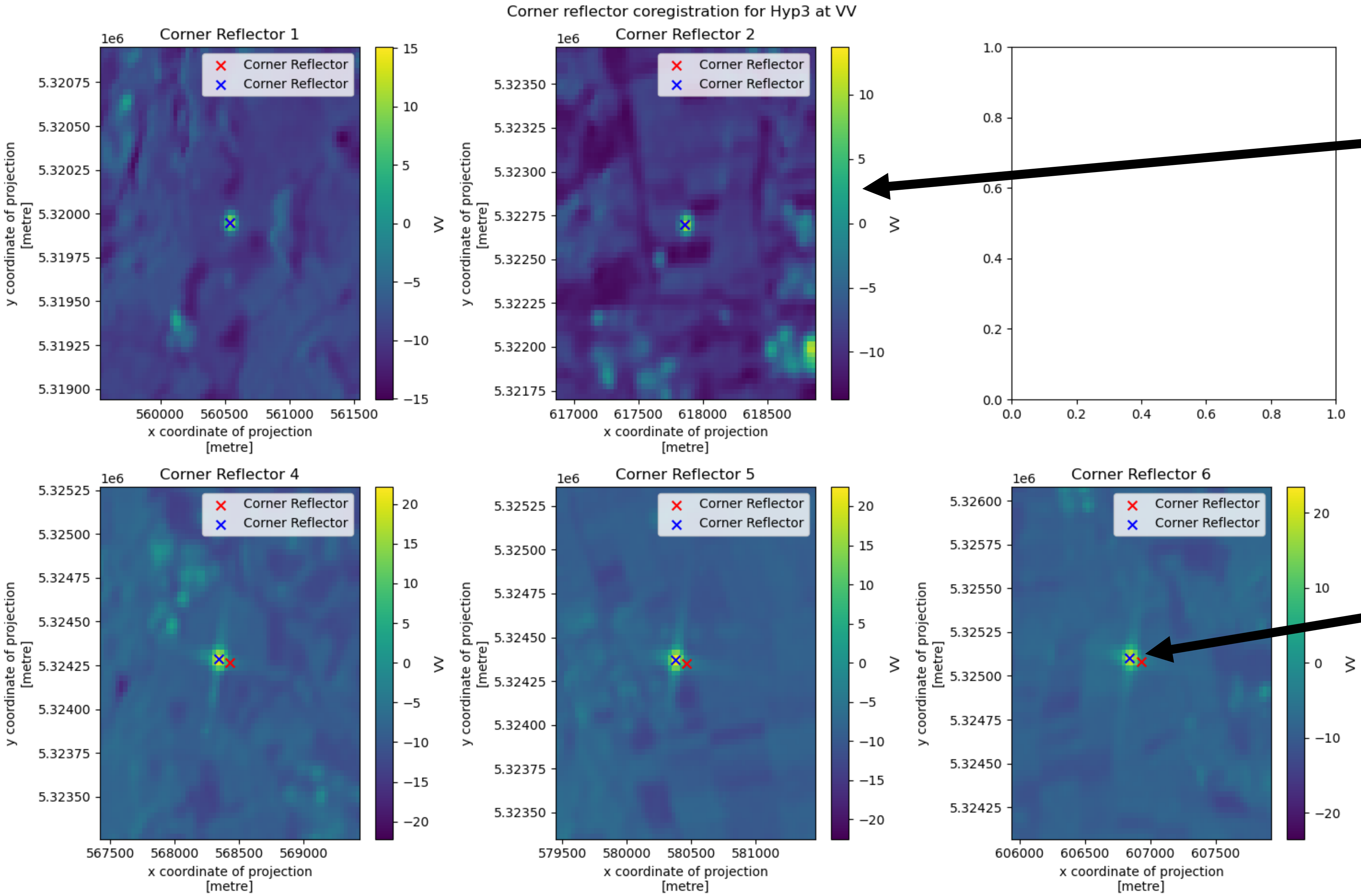
- Hyp3 – CPU times: user 50.6 s, sys: 1min 11s, total: 2min 1s Wall time: 1h 21min 2s

- Opera - CPU times: user 5.93 s, sys: 5.49 s, total: 11.4 s Wall time: 7min 31s

- Microsoft Stac - CPU times: user 7.53 s, sys: 2.19 s, total: 9.72 s Wall time: 30.1 s

Hyp3 Corner Reflector Results

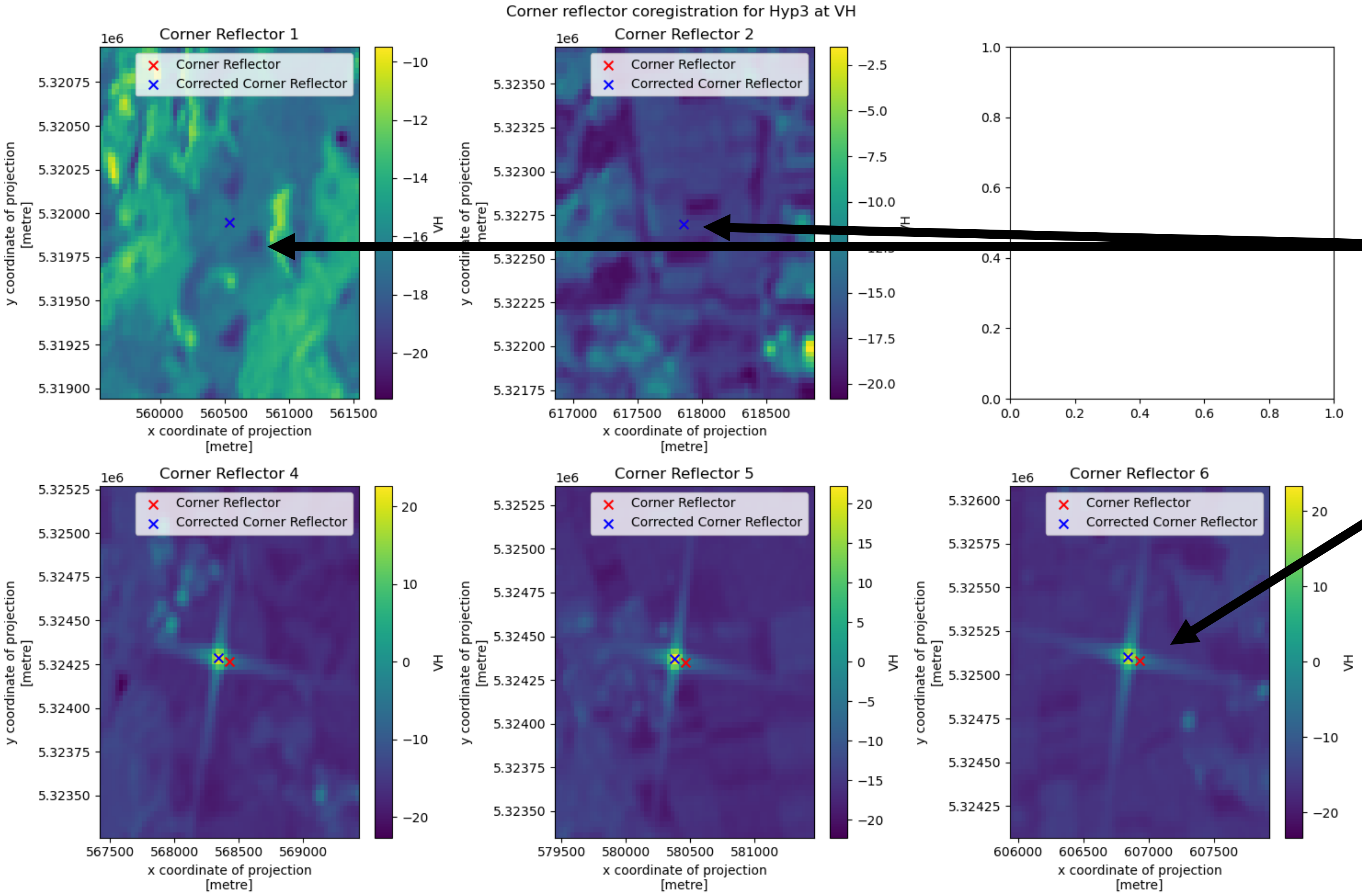
Hyp3 VV coregistration on corner reflectors



Great positional accuracy for the passive targets (1, 2)

These offsets for the transponders (4 -6) are present in all datasets.

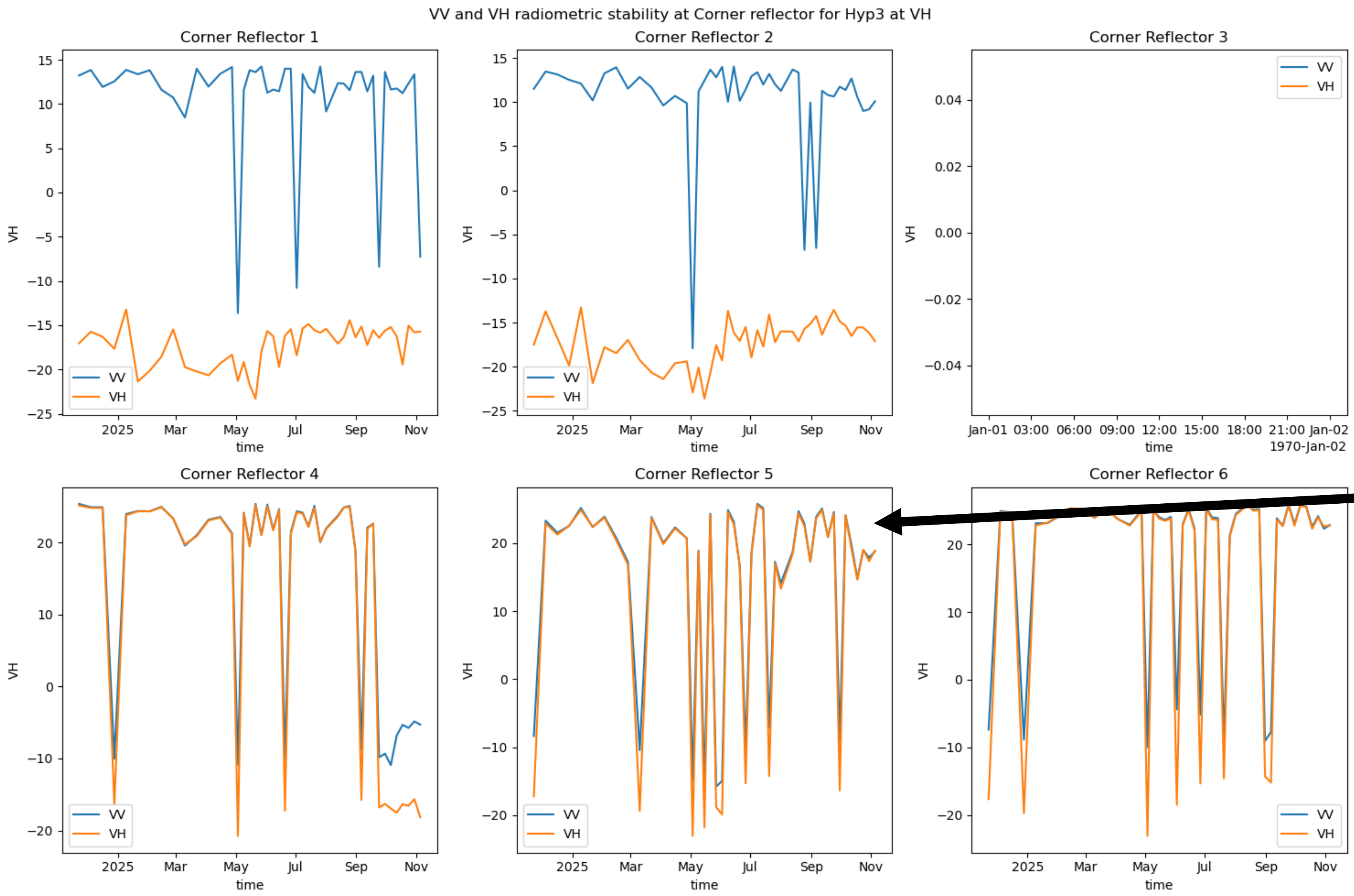
Hyp3 VH coregistration on corner reflectors



No VH response for passive targets (1, 2) as expected

Red x is reported location. Blue x is my "corrected coordinates" used to extract timeseries in following figure. Same errors seen in Opera. Maybe projection error or transponder differences?

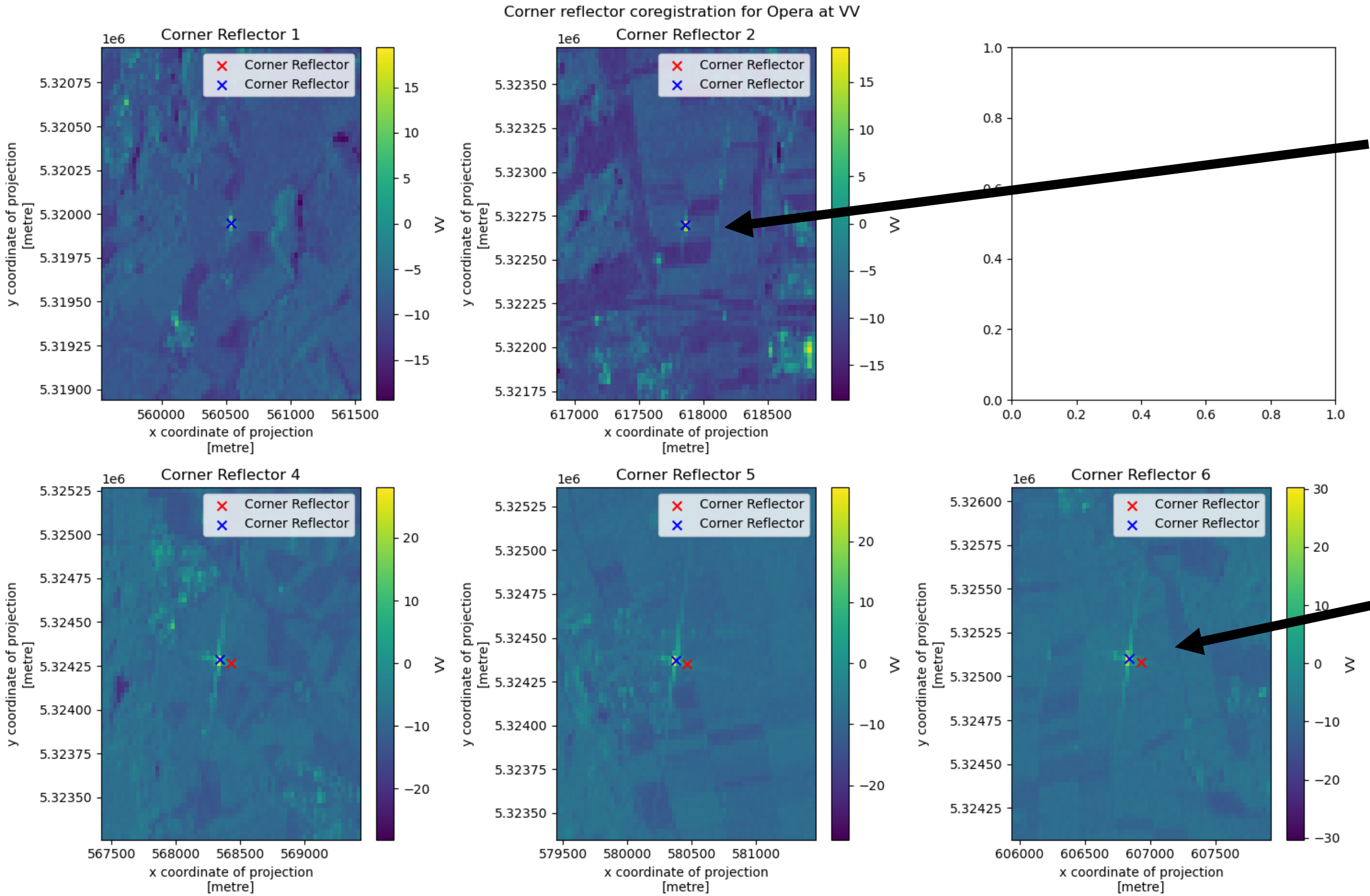
Hyp3 Corner reflector radiometric stability



Even when the transponders are on we see 1-2 dB variations

Opera Corner Reflector Results

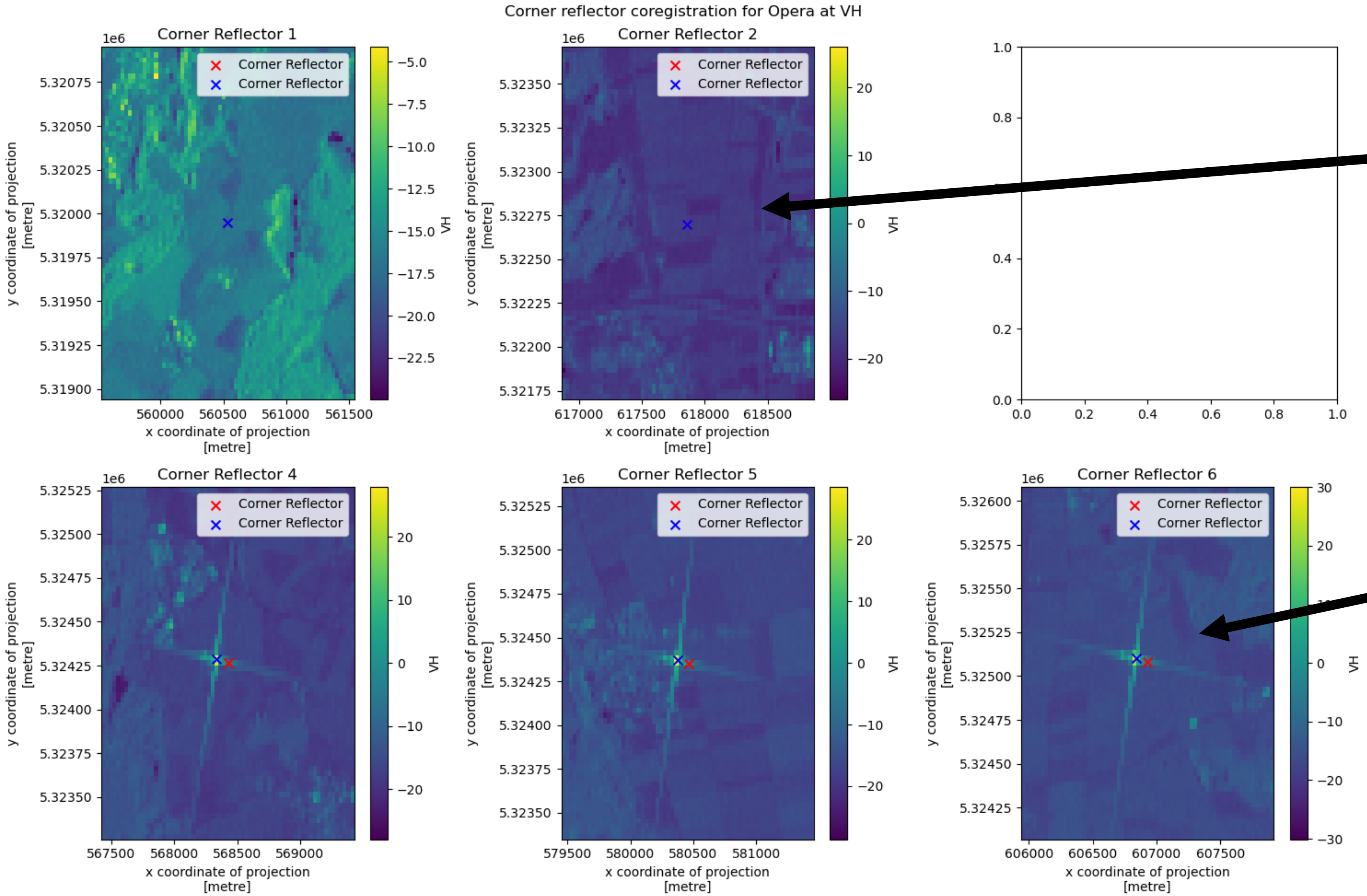
Opera VV coregistration on corner reflectors



Good location matches on passive CRs

Same errors seen in Hyp3. Maybe projection error or transponder differences?

Opera VH coregistration on corner reflectors

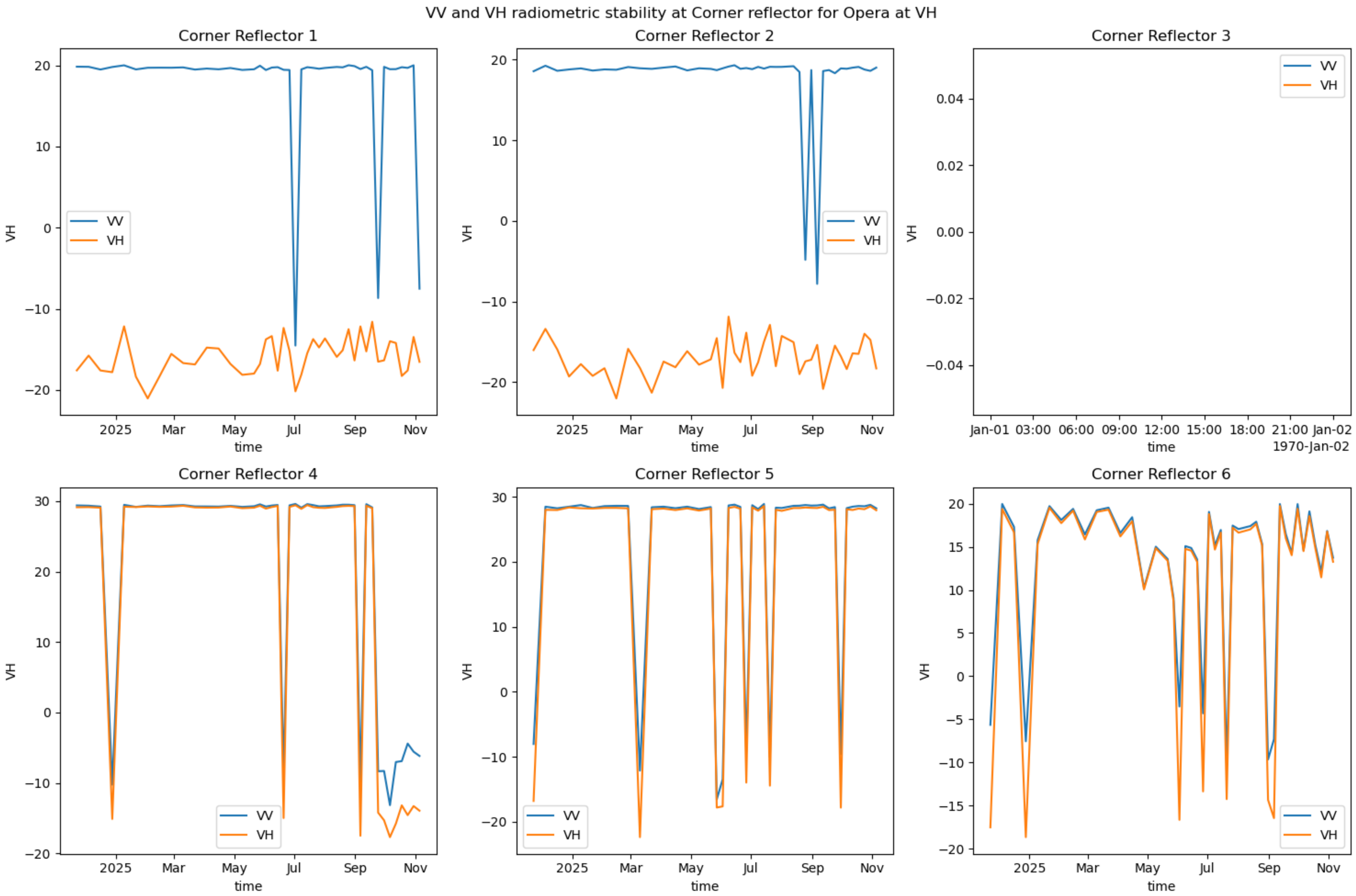


No VH response for passive targets as expected

Same errors seen in Hyp3. Maybe projection error or transponder differences?

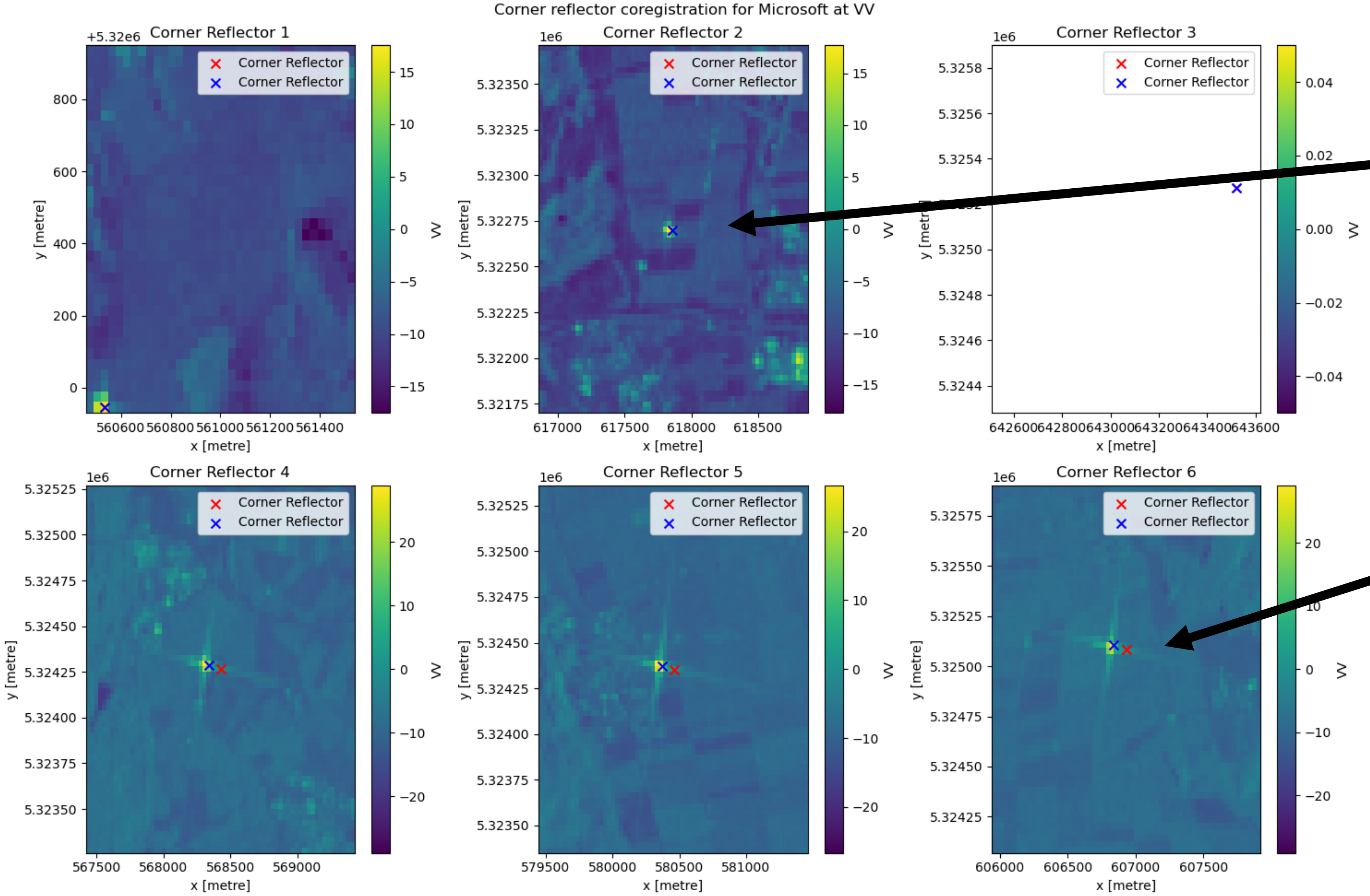
Opera corner reflector radiometric stability

Overall great radiometric accuracy. Most drops are periods transponders are off or passive targets are in “stored” configuration



Microsoft Stac Corner Reflector Results

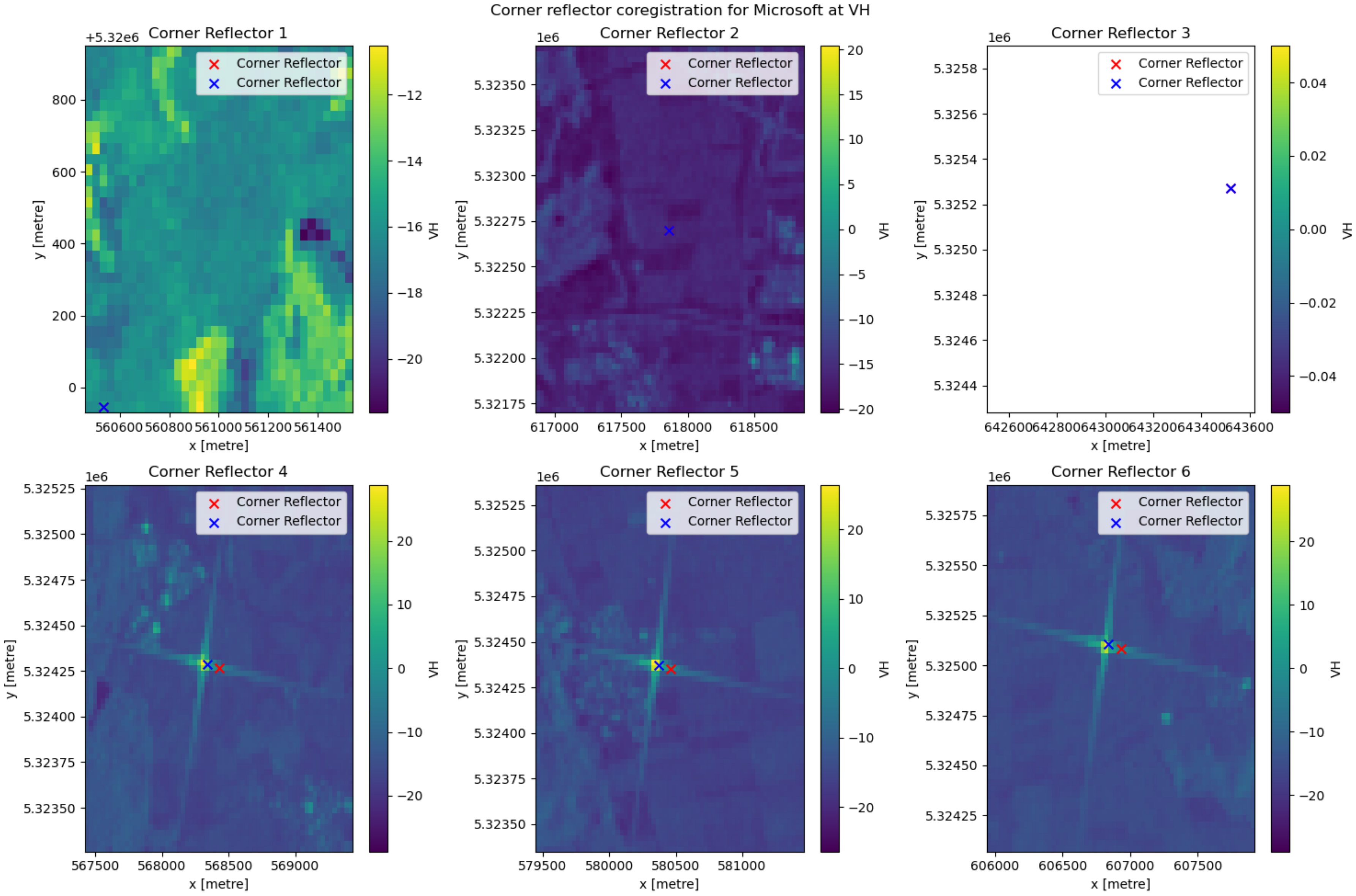
Microsoft Stac VV coregistration on corner reflectors



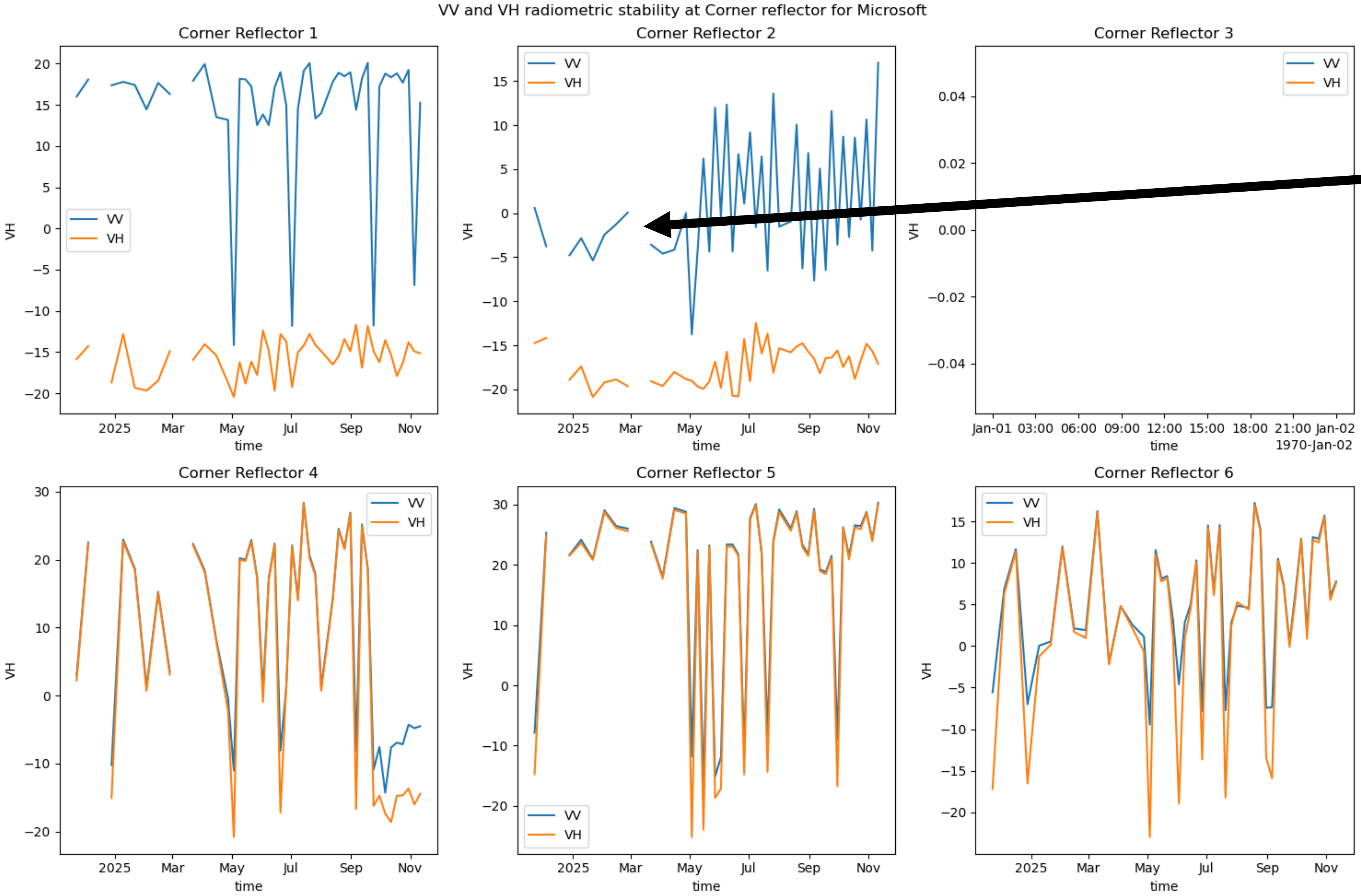
Larger positional errors for passive targets

Same errors as hyp3 and opera

Microsoft Stac VH coregistration on corner reflectors



Microsoft Stac corner reflector radiometric stability



These gaps show that some of the Microsoft images are nans in regions that are not in opera/hyp3

Worse radiometric consistency for transponders

Conclusions

- Microsoft stac is the obvious choice for speed (30 seconds vs 7 minutes and 120 minutes) and picking up most recent (day of overpass) images
- But had some non-sensical dates and bad radiometric consistency and worst coregistration
- Best radiometric consistency and coregistration found with Opera products.
- But missed 2 dates from asf-search setup...